Scott's Coherence Functional

A Nonlinear Trajectory-Dependent Framework for Damping, Oscillatory Memory, and Higher-Order Dynamical Interactions

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November 2026

Abstract

This paper introduces Scott's Coherence Functional, a multicomponent mathematical construction designed to investigate interactions among nonlinear state dependence, accumulated temporal effects, exponential attenuation, oscillatory phase, higher-order derivatives, and acceleration-dependent contributions.

The proposed functional is formulated through three principal components: a nonlocal integral contribution involving mixed second-order spatial derivatives, a product-weighted third-order derivative contribution, and a nonlinear correction incorporating velocity and acceleration.

The construction is examined within an appropriate space of twice continuously differentiable trajectories. Under sufficient regularity, boundedness, and sign assumptions, we establish well-definedness, continuity in time, and an explicit global estimate on finite intervals.

We further investigate the mathematical structure of the integral kernel, derive estimates for exponentially attenuated memory contributions, and examine parameter sensitivity with respect to generalized damping coefficients.

Additional results concern perturbations of trajectories, limiting regimes, nonlinear attenuation, and possible extensions to higher-order dynamical systems.

The analysis also identifies limitations of the current construction, particularly concerning trajectory-independent bounds, variational differentiability, conservation properties, and numerical convergence.

The purpose of this work is to establish a mathematically explicit framework for examining the structural properties of Scott's Coherence Functional and to provide a foundation for subsequent analytical investigation.

Keywords: coherence functional; nonlinear dynamics; functional analysis; oscillatory integrals; exponential damping; memory kernels; trajectory perturbation; asymptotic analysis; higher-order derivatives.

2020 Mathematics Subject Classification: Primary 26B15; Secondary 34A34, 45J05, 47H30.

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1 Introduction

Many mathematical systems involve interactions between multiple variables that evolve continuously over time.

Such systems may contain several different mechanisms simultaneously, including nonlinear coupling, accumulated phase, exponential attenuation, state-dependent weighting, and higher-order derivatives.

A mathematical representation of these interactions must account for both the instantaneous configuration of a system and the influence of its evolution over an interval of time.

The purpose of this paper is to introduce a trajectory-dependent functional that combines these mechanisms into a single mathematical construction.

We refer to this construction as *Scott's Coherence Functional*.

The functional is designed around three mathematically distinguishable contributions.

The first is a nonlocal integral term combining mixed second-order spatial derivatives, exponential damping, accumulated oscillatory phase, and nonlinear velocity dependence.

The second introduces third-order derivatives weighted by products of state-dependent factors.

The third introduces a nonlinear correction involving both velocity and acceleration.

Together, these contributions provide a framework for investigating the relationships between instantaneous state information, historical trajectory dependence, and higher-order dynamical quantities.

1.1 Motivation

A purely instantaneous mathematical expression may be insufficient when the behavior of a system depends on its preceding evolution.

Integral expressions provide one way of representing such dependence.

In particular, an integral whose upper limit varies with observation time can incorporate information from earlier states while allowing the influence of those states to depend on subsequent evolution.

Similarly, exponential factors can represent attenuation, while trigonometric factors can represent oscillatory interactions.

Nonlinear denominators provide additional mechanisms for modifying the relative influence of state variables, velocities, and accelerations.

Scott's Coherence Functional combines these ideas into a single expression.

1.2 Principal Mathematical Questions

The construction raises several questions:

1. Under what regularity conditions is the functional well-defined?
2. Can its magnitude be bounded uniformly over a finite observation interval?
3. How does accumulated damping influence historical trajectory contributions?
4. What role does oscillatory phase play in the resulting integral?
5. Is the functional continuous with respect to time and perturbations of the trajectory?
6. How sensitive is the construction to changes in its defining parameters?
7. What limiting behavior occurs under strong damping, weak damping, and short-time conditions?
8. Can the construction be generalized to higher-order or variational mathematical systems?

The following sections develop the mathematical framework required to investigate these questions.

2 Definition and Mathematical Framework

2.1 State trajectories

Let $n \in \mathbb{N}$ be a positive integer and let $T > 0$ be a finite observation horizon.

Consider a trajectory

$$\mathbf{x}(t) = (x_1(t), x_2(t), \ldots, x_n(t))$$

taking values in $\mathbb{R}^n$.

Initially, assume that

$$\mathbf{x} \in C^2([0,T], \mathbb{R}^n).$$

The first and second derivatives are denoted by

$$\dot{x}_i(t) = \frac{\mathrm{d}x_i}{\mathrm{d}t},$$

and

$$\ddot{x}_i(t) = \frac{\mathrm{d}^2 x_i}{\mathrm{d}t^2}.$$

The functions Φ_i and Ψ_i describe state-dependent quantities.

The functions ω_i and Ω_{ijk} determine damping and oscillatory contributions.

The function Θ_{ijk} represents an additional accumulated phase contribution.

The coefficients χ_m, κ_{ij}, and Λ_{ijk} determine the strength of the corresponding nonlinear terms.

2.2 Functional spaces

For a fixed interval $[0,T]$, define

$$\mathcal{X}_T = C^2([0,T], \mathbb{R}^n).$$

Equip this space with the norm

$$\begin{aligned} \|\mathbf{x}\|_{C^2} = & \max_{t\in[0,T]} \|\mathbf{x}(t)\| \\ & + \max_{t\in[0,T]} \|\dot{\mathbf{x}}(t)\| \\ & + \max_{t\in[0,T]} \|\ddot{\mathbf{x}}(t)\| . \end{aligned}$$

With this norm, $\mathcal{X}_T$ is a Banach space.

This provides a natural setting for studying trajectory-dependent mathematical quantities because the norm simultaneously controls the state, velocity, and acceleration.

Assume that

$$\Phi_i : \mathbb{R}^n \times [0, T] \to \mathbb{R}$$

and

$$\Psi_i : \mathbb{R}^n \times [0, T] \to \mathbb{R}.$$

All spatial derivatives appearing below are evaluated at the corresponding trajectory point.

For example,

$$\frac{\partial^2 \Phi_i}{\partial x_j \partial x_k}$$

inside the integral denotes

$$\frac{\partial^2 \Phi_i}{\partial x_j \partial x_k}(\mathbf{x}(s), s).$$

Likewise, the mixed derivative in the second component is evaluated as

$$\frac{\partial^3 \Phi_i}{\partial x_i^2 \partial t}(\mathbf{x}(t), t),$$

where the partial derivative with respect to time is taken while holding the spatial argument fixed.

2.3 Definition of the functional

Definition 2.1 (Scott's Coherence Functional)**.** Let $T > 0$, let $n \in \mathbb{N}$, and let

$$\mathbf{x} \in \mathcal{X}_T.$$

Whenever all required derivatives and integrals exist, define

$$\mathfrak{Q} : \mathcal{X}_T \times [0,T] \to \mathbb{R}$$

by the expression

$$\begin{aligned}
\mathfrak{Q}(\mathbf{x},t) = \sum_{i,j,k=1}^{n} \int_0^t & \frac{\partial^2 \Phi_i}{\partial x_j \partial x_k}(\mathbf{x}(s),s) \\
& \cdot \exp\left(-\int_s^t \frac{\omega_i(\mathbf{x}(\xi))}{1+|\dot{x}_j(\xi)|}\ \mathrm{d}\xi\right) \\
& \cdot \sin\left(\int_s^t \Omega_{ijk}(\mathbf{x}(\eta))\ \mathrm{d}\eta + \Theta_{ijk}(s,t)\right) \\
& \cdot \frac{\dot{x}_j(s)\dot{x}_k(s)}{1+\sum_{m=1}^{n} \chi_m x_m^2(s)} \\
& \cdot \left(1 + \frac{\partial \Psi_i/\partial x_j(\mathbf{x}(s),s)}{1+|\partial \Psi_k/\partial x_i(\mathbf{x}(s),s)|}\right) \mathrm{d}s \\
+ \sum_{i=1}^{n} & \frac{\partial^3 \Phi_i}{\partial x_i^2 \partial t}(\mathbf{x}(t),t) \prod_{j=1}^{n}(1+\kappa_{ij}x_j^2(t)) \\
+ \sum_{i,j,k=1}^{n} & \Lambda_{ijk} \frac{\dot{x}_i(t)\dot{x}_j(t)\ddot{x}_k(t)}{1+|\dot{x}_i(t)\dot{x}_j(t)|+|\ddot{x}_k(t)|}.
\end{aligned} \tag{2.1}$$

The resulting quantity is called Scott's Coherence Functional.

The functional can be decomposed into three principal components:

$$\mathfrak{Q} = \mathfrak{Q}_I + \mathfrak{Q}_D + \mathfrak{Q}_A.$$

Here:

- $\mathfrak{Q}_I$ denotes the nonlocal integral contribution. - $\mathfrak{Q}_D$ denotes the product-weighted derivative contribution. - $\mathfrak{Q}_A$ denotes the nonlinear acceleration contribution.

This decomposition will be used throughout the subsequent analysis.

2.4 The integral component

Define

$$\mathfrak{Q}_I(\mathbf{x},t) = \sum_{i,j,k=1}^{n} \int_0^t H_{ijk}(\mathbf{x},s) \cdot E_{ij}(s,t)\sin(P_{ijk}(s,t))\ \mathrm{d}s,$$

where the instantaneous trajectory-dependent factor is

$$H_{ijk}(\mathbf{x},s) = \frac{\partial^2 \Phi_i}{\partial x_j \partial x_k}(\mathbf{x}(s),s) \cdot \frac{\dot{x}_j(s)\dot{x}_k(s)}{1+\sum_{m=1}^{n}\chi_m x_m^2(s)} \cdot \left(1+\frac{\partial \Psi_i/\partial x_j(\mathbf{x}(s),s)}{1+|\partial \Psi_k/\partial x_i(\mathbf{x}(s),s)|}\right).$$

The exponential attenuation factor is

$$E_{ij}(s,t) = \exp\left(-\int_s^t \frac{\omega_i(\mathbf{x}(\xi))}{1+|\dot{x}_j(\xi)|}\ \mathrm{d}\xi\right).$$

The accumulated oscillatory phase is

$$P_{ijk}(s,t) = \int_s^t \Omega_{ijk}(\mathbf{x}(\eta))\ \mathrm{d}\eta + \Theta_{ijk}(s,t).$$

This decomposition separates the instantaneous trajectory contribution from the historical weighting mechanism.

2.5 The derivative component

The second component is defined by

$$\mathfrak{Q}_D(\mathbf{x},t) = \sum_{i=1}^{n} \frac{\partial^3 \Phi_i}{\partial x_i^2 \partial t}(\mathbf{x}(t),t) \prod_{j=1}^{n}(1+\kappa_{ij}x_j^2(t)).$$

It introduces higher-order derivative information combined with nonlinear state-dependent weighting.

2.6 The acceleration component

The final component is

$$\mathfrak{Q}_A(\mathbf{x},t) = \sum_{i,j,k=1}^{n} \Lambda_{ijk} \frac{\dot{x}_i(t)\dot{x}_j(t)\ddot{x}_k(t)}{1+|\dot{x}_i(t)\dot{x}_j(t)|+|\ddot{x}_k(t)|}.$$

This component introduces nonlinear dependence on both velocity and acceleration.

3 Assumptions and Preliminary Results

A rigorous analysis requires sufficient conditions on the coefficient functions, trajectory, and parameters.

The following assumptions will be used throughout the principal existence and boundedness results.

Assumption 3.1 (Regularity)**.** The functions Φ_i possess continuous spatial derivatives through order three.

Their mixed derivatives

$$\frac{\partial^3 \Phi_i}{\partial x_i^2 \partial t}$$

are continuous.

The functions Ψ_i, ω_i, and Ω_{ijk} are continuous wherever evaluated.

The phase functions Θ_{ijk} are continuous on

$$\Delta_T = \{(s,t) \in \mathbb{R}^2 : 0 \le s \le t \le T\}.$$

Assumption 3.2 (Nonnegative state weights)**.** The coefficients satisfy

$$\chi_m \ge 0, \qquad \kappa_{ij} \ge 0.$$

All coefficients Λ_{ijk} are finite real numbers.

Assumption 3.3 (Nonnegative damping)**.** For every relevant state,

$$\omega_i(\mathbf{x}) \geq 0.$$

Assumption 3.4 (Trajectory regularity)**.** The trajectory satisfies

$$\mathbf{x} \in C^2([0,T], \mathbb{R}^n).$$

Remark 3.5. These assumptions are sufficient conditions rather than necessary conditions.

In particular, nonnegative damping guarantees that the exponential factor does not exceed one in absolute value.

If negative damping is permitted, the exponential factor may instead produce amplification.

3.1 Bounded trajectory quantities

Since the trajectory and its derivatives are continuous on a compact interval, the following quantities are finite:

$$X = \max_{t\in[0,T]} \|\mathbf{x}(t)\|,$$

$$V = \max_{\substack{1\leq i\leq n \\ t\in[0,T]}} |\dot{x}_i(t)|,$$

and

$$A = \max_{\substack{1\leq i\leq n \\ t\in[0,T]}} |\ddot{x}_i(t)|.$$

These constants will be used in the subsequent estimates.

3.2 Auxiliary coefficient bounds

Define

$$M_2 = \max_{i,j,k} \sup_{0\leq s\leq T} \left| \frac{\partial^2 \Phi_i}{\partial x_j \partial x_k}(\mathbf{x}(s), s) \right|.$$

Similarly, define

$$P = \max_{i,j} \sup_{0 \le s \le T} \left| \frac{\partial \Psi_i}{\partial x_j}(\mathbf{x}(s), s) \right|.$$

For the third-order derivative contribution, let

$$M_3 = \max_{i} \sup_{0 \le t \le T} \left| \frac{\partial^3 \Phi_i}{\partial x_i^2 \partial t}(\mathbf{x}(t), t) \right|.$$

Finally, define

$$K = \max_{i,j} \kappa_{ij}, \qquad L = \max_{i,j,k} |\Lambda_{ijk}|.$$

All these constants are finite under the stated assumptions.

4 Existence and Well-Definedness

Theorem 4.1 (Existence on Finite Time Intervals)**.** *Suppose the regularity, nonnegative-weight, nonnegative-damping, and trajectory assumptions hold.*

Then Scott's Coherence Functional is finite and well-defined for every

$$t \in [0, T].$$

Proof. Consider first the integral component.

The trajectory is continuous, and the coefficient functions have the required continuity properties.

Consequently, the mixed derivative factor

$$\frac{\partial^2 \Phi_i}{\partial x_j \partial x_k}(\mathbf{x}(s), s)$$

is continuous.

The denominator

$$1 + \sum_{m=1}^{n} \chi_m x_m^2(s)$$

is at least one because

$$\chi_m \geq 0.$$

Likewise,

$$1 + \left| \frac{\partial \Psi_k}{\partial x_i} \right| \geq 1.$$

Thus, neither denominator can vanish.

The damping integral is finite because its integrand is continuous on a compact interval.

Therefore, the exponential factor is well-defined and finite.

The phase integral is also finite, and the sine function is defined for every real phase argument.

It follows that the complete integrand is continuous and bounded on the relevant integration interval.

Its integral therefore exists and is finite.

Next, consider the derivative component.

The mixed third-order derivatives are continuous, and the trajectory is bounded.

Since the coefficients κ_{ij} are finite, every factor

$$1 + \kappa_{ij} x_j^2(t)$$

is finite.

The finite product and finite summation therefore produce a finite quantity.

Finally, consider the acceleration component.

Its denominator satisfies

$$1 + |\dot{x}_i \dot{x}_j| + |\ddot{x}_k| \geq 1.$$

The numerator is bounded because velocity and acceleration are bounded on $[0, T]$.

All coefficients Λ_{ijk} are finite, and the sums contain finitely many terms.

Thus, the acceleration component is also finite.

All three components are therefore well-defined, proving the theorem.

□

4.1 A bound for the integral component

The integral component can be bounded explicitly.

First, nonnegative damping gives

$$0 < E_{ij}(s,t) \leq 1.$$

Furthermore,

$$|\sin(P_{ijk}(s,t))| \leq 1.$$

The state-dependent denominator satisfies

$$1 + \sum_{m=1}^{n} \chi_m x_m^2(s) \geq 1.$$

The derivative ratio satisfies

$$\left| 1 + \frac{\partial \Psi_i / \partial x_j}{1 + |\partial \Psi_k / \partial x_i|} \right| \leq 1 + P.$$

Consequently,

$$|H_{ijk}(\mathbf{x}, s)| \leq M_2 V^2 (1 + P).$$

Proposition 4.2 (Integral Component Bound)**.** *Under the preceding assumptions,*

$$|\mathfrak{Q}_I(\mathbf{x}, t)| \leq n^3 T M_2 V^2 (1 + P).$$

Proof. Applying the triangle inequality gives

$$|\mathfrak{Q}_I(\mathbf{x},t)| \leq \sum_{i,j,k=1}^{n} \int_0^t |H_{ijk}(\mathbf{x},s)| \cdot E_{ij}(s,t)|\sin(P_{ijk}(s,t))|\ \mathrm{d}s.$$

Using the established estimates,

$$\begin{aligned} |\mathfrak{Q}_I(\mathbf{x},t)| &\leq \sum_{i,j,k=1}^{n} \int_0^t M_2 V^2 (1+P)\ \mathrm{d}s \\ &\leq n^3 T M_2 V^2 (1+P). \end{aligned}$$

This proves the proposition.

□

5 Global Bounds for the Functional

We now establish bounds for the derivative and acceleration contributions.

5.1 Derivative component estimate

Since

$$|x_j(t)| \leq X$$

and

$$\kappa_{ij} \leq K,$$

we have

$$1 + \kappa_{ij} x_j^2(t) \leq 1 + KX^2.$$

Therefore,

$$\prod_{j=1}^{n}(1+\kappa_{ij}x_j^2(t)) \le (1+KX^2)^n.$$

It follows that

$$|\mathfrak{Q}_D(\mathbf{x},t)| \le nM_3(1+KX^2)^n.$$

5.2 Acceleration component estimate

The denominator satisfies

$$1+|\dot{x}_i\dot{x}_j|+|\ddot{x}_k| \ge 1.$$

Hence,

$$\left|\frac{\dot{x}_i\dot{x}_j\ddot{x}_k}{1+|\dot{x}_i\dot{x}_j|+|\ddot{x}_k|}\right| \le V^2A.$$

Since

$$|\Lambda_{ijk}| \le L,$$

we obtain

$$|\mathfrak{Q}_A(\mathbf{x},t)| \le n^3LV^2A.$$

5.3 Main global boundedness theorem

Theorem 5.1 (Global Bound on a Finite Interval)**.** *Under the assumptions of this paper,*

$$\begin{aligned}|\mathfrak{Q}(\mathbf{x},t)| \le\ & n^3TM_2V^2(1+P)\\ & + nM_3(1+KX^2)^n\\ & + n^3LV^2A.\end{aligned}$$

for every $t \in [0, T]$.

Proof. The decomposition

$$\mathfrak{Q} = \mathfrak{Q}_I + \mathfrak{Q}_D + \mathfrak{Q}_A$$

and the triangle inequality yield

$$|\mathfrak{Q}| \leq |\mathfrak{Q}_I| + |\mathfrak{Q}_D| + |\mathfrak{Q}_A|.$$

Substituting the three previously established component bounds gives

$$\begin{aligned} |\mathfrak{Q}| \leq\ & n^3 T M_2 V^2 (1 + P) \\ & + n M_3 (1 + K X^2)^n \\ & + n^3 L V^2 A. \end{aligned}$$

This proves the theorem.

□

Remark 5.2. The estimate depends on the trajectory through X, V, and A.

It is therefore a finite-interval estimate rather than a universal bound independent of the trajectory.

Additional assumptions would be necessary to obtain a trajectory-independent estimate.

6 The Damping and Oscillatory Components

6.1 Exponential damping

The exponential factor is

$$E_{ij}(s, t) = \exp\left(- \int_s^t \frac{\omega_i(\mathbf{x}(\xi))}{1 + |\dot{x}_j(\xi)|} \, \mathrm{d}\xi \right).$$

This factor introduces accumulated attenuation over the interval from s to t.

Because the damping function is nonnegative,

$$0 < E_{ij}(s,t) \leq 1.$$

The velocity-dependent denominator modifies the effective damping rate.

In particular, define

$$r_{ij}(\xi) = \frac{\omega_i(\mathbf{x}(\xi))}{1 + |\dot{x}_j(\xi)|}.$$

For fixed nonnegative ω_i, increasing the velocity magnitude decreases this rate.

Proposition 6.1 (Damping Monotonicity)**.** *For every $a \geq 0$, the function*

$$f(v) = \frac{a}{1+v}, \qquad v \geq 0,$$

is nonincreasing.

Proof. Differentiation gives

$$f'(v) = -\frac{a}{(1+v)^2} \leq 0.$$

Therefore, increasing v cannot increase the effective damping rate.

□

6.2 Accumulated oscillatory phase

The oscillatory factor is

$$\sin\left(\int_s^t \Omega_{ijk}(\mathbf{x}(\eta))\ \mathrm{d}\eta + \Theta_{ijk}(s,t)\right).$$

Its argument contains two contributions.

The first is the accumulated trajectory-dependent phase.

The second is the additional phase Θ_{ijk}.

The combination permits oscillatory behavior to depend on both the evolving state and an independent phase contribution.

Since

$$-1 \leq \sin(\theta) \leq 1,$$

the oscillatory factor cannot increase the absolute value of the integrand beyond the remaining multiplicative factors.

Remark 6.2. The phase contribution can change the sign of individual portions of the integral.

Consequently, different regions of the integration interval may reinforce or partially cancel one another.

This makes the integral component sensitive to phase relationships along the trajectory.

7 Nonlinear State and Acceleration Contributions

7.1 State-dependent attenuation

The integral component contains the factor

$$R(\mathbf{x}, s) = \frac{1}{1 + \sum_{m=1}^{n} \chi_m x_m^2(s)}.$$

Under the nonnegative-weight assumption,

$$0 < R(\mathbf{x}, s) \leq 1.$$

For fixed velocities, increasing the weighted state magnitude decreases this factor.

Thus, the state-dependent denominator introduces a nonlinear attenuation mechanism.

7.2 Product-weighted derivative contribution

The derivative component contains

$$W_i(\mathbf{x}, t) = \prod_{j=1}^{n} (1 + \kappa_{ij} x_j^2(t)).$$

Since every coefficient is nonnegative,

$$W_i(\mathbf{x}, t) \geq 1.$$

Unlike the denominator in the integral component, this product increases with the magnitude of the weighted state coordinates.

It therefore acts as a multiplicative state-dependent weighting mechanism.

7.3 Acceleration correction

Consider

$$\mathfrak{Q}_A = \sum_{i,j,k=1}^{n} \Lambda_{ijk} \frac{\dot{x}_i \dot{x}_j \ddot{x}_k}{1 + |\dot{x}_i \dot{x}_j| + |\ddot{x}_k|}.$$

For convenience, define

$$u = \dot{x}_i \dot{x}_j, \qquad v = \ddot{x}_k.$$

Then

$$\left| \frac{uv}{1 + |u| + |v|} \right| = \frac{|u||v|}{1 + |u| + |v|}.$$

Because

$$1 + |u| + |v| \geq |u|$$

and

$$1 + |u| + |v| \geq |v|,$$

we obtain

$$\left|\frac{uv}{1+|u|+|v|}\right| \leq \min(|u|,|v|).$$

Thus,

$$\left|\frac{\dot{x}_i\dot{x}_j\ddot{x}_k}{1+|\dot{x}_i\dot{x}_j|+|\ddot{x}_k|}\right| \leq \min\left(|\dot{x}_i\dot{x}_j|,|\ddot{x}_k|\right).$$

This inequality provides an additional estimate for the acceleration contribution.

7.4 Structural interpretation

The acceleration term combines two velocity factors with one acceleration factor.

Its denominator guarantees that the expression remains well-defined for all real velocity and acceleration values.

It also moderates the growth of the numerator when the quantities become large.

However, the denominator does not imply that the entire acceleration contribution is universally bounded independently of the trajectory.

Such a conclusion would require further restrictions on the admissible velocities and accelerations.

8 Continuity and Time Dependence

The preceding results establish existence and boundedness.

We now examine continuity.

Theorem 8.1 (Continuity in Time)**.** *Suppose the regularity assumptions hold and all coefficient functions are continuous along the trajectory.*

Then

$$t \longmapsto \mathfrak{Q}(\mathbf{x},t)$$

is continuous on $[0, T]$.

Proof. Consider the integral component.

Its integrand is continuous in the variables s and t on the compact triangular domain

$$\Delta_T = \{(s, t) : 0 \leq s \leq t \leq T\}.$$

The exponential and sine functions are continuous compositions of the accumulated integrals.

All denominators remain strictly positive.

The parameter-dependent integral therefore varies continuously with its upper limit and with its integrand.

Hence, every summand in the integral component is continuous in t.

The derivative component is continuous because its derivatives and state-dependent factors are continuous.

The acceleration component is continuous because the trajectory belongs to C^2 and its denominator is strictly positive.

Since the functional is a finite sum of these continuous components, it is continuous on $[0, T]$.

□

Corollary 8.2 (Attainment of Extreme Values)**.** *The functional attains a maximum and a minimum on every compact interval* $[0, T]$.

Proof. The functional is continuous on the compact interval $[0, T]$.

The extreme value theorem therefore guarantees that it attains both a maximum and a minimum.

□

9 Nonlocal Structure and Memory Dependence

An important feature of the integral component is that its value at time t depends on trajectory information accumulated over the interval preceding t.

This property distinguishes the integral component from purely instantaneous nonlinear expressions.

9.1 Construction of the memory kernel

Recall the definitions

$$D_{ij}(s,t) = \int_s^t \frac{\omega_i(\mathbf{x}(\xi))}{1+|\dot{x}_j(\xi)|}\ \mathrm{d}\xi$$

and

$$P_{ijk}(s,t) = \int_s^t \Omega_{ijk}(\mathbf{x}(\eta))\ \mathrm{d}\eta + \Theta_{ijk}(s,t).$$

Define the combined memory kernel by

$$K_{ijk}(s,t) = e^{-D_{ij}(s,t)} \sin(P_{ijk}(s,t)).$$

The integral component can consequently be represented as

$$\mathfrak{Q}_I(\mathbf{x},t) = \sum_{i,j,k=1}^{n} \int_0^t K_{ijk}(s,t) H_{ijk}(\mathbf{x},s)\ \mathrm{d}s.$$

This representation separates the historical weighting mechanism from the instantaneous trajectory-dependent contribution.

9.2 Kernel magnitude estimate

Under nonnegative damping,

$$D_{ij}(s,t) \geq 0.$$

Therefore,

$$\begin{aligned}|K_{ijk}(s,t)| &= e^{-D_{ij}(s,t)}|\sin(P_{ijk}(s,t))| \\ &\leq e^{-D_{ij}(s,t)} \\ &\leq 1.\end{aligned}$$

Thus,

$$|K_{ijk}(s,t)| \leq 1.$$

9.3 Exponential memory attenuation

Suppose there exists a constant $\delta > 0$ such that

$$\frac{\omega_i(\mathbf{x}(\xi))}{1 + |\dot{x}_j(\xi)|} \geq \delta$$

throughout the relevant interval.

Then

$$D_{ij}(s,t) \geq \delta(t-s).$$

Consequently,

$$|K_{ijk}(s,t)| \leq e^{-\delta(t-s)}.$$

This establishes an exponentially decaying upper bound on the influence of earlier trajectory states.

9.4 Uniform memory-weight estimate

Integrating the preceding inequality gives

$$\int_0^t |K_{ijk}(s,t)| \, \mathrm{d}s \leq \int_0^t e^{-\delta(t-s)} \, \mathrm{d}s.$$

Evaluating the integral,

$$\int_0^t e^{-\delta(t-s)}\,\mathrm{d}s = \frac{1-e^{-\delta t}}{\delta}.$$

Therefore,

$$\int_0^t |K_{ijk}(s,t)|\,\mathrm{d}s \le \frac{1}{\delta}.$$

The bound is independent of the observation time t.

This demonstrates that a uniformly positive effective damping rate produces a uniformly bounded total memory weight, even as the observation interval increases.

Remark 9.1. The estimate controls the total magnitude of the memory kernel.

It does not imply that every individual historical contribution is positive, since the oscillatory phase may change its sign.

10 Parameter Sensitivity and Perturbation Analysis

The behavior of Scott's Coherence Functional depends on several independent parameters.

These include damping coefficients, oscillatory coefficients, nonlinear state weights, and acceleration coefficients.

A systematic analysis of their influence provides additional information concerning the structural sensitivity of the construction.

10.1 Generalized damping parameter

Introduce a nonnegative parameter α into the damping expression:

$$E_{ij}^{(\alpha)}(s,t) = \exp(-\alpha D_{ij}(s,t)).$$

For $\alpha = 1$, the original damping factor is recovered.

Differentiating with respect to α,

$$\frac{\partial E_{ij}^{(\alpha)}}{\partial \alpha} = -D_{ij}(s,t)e^{-\alpha D_{ij}(s,t)}.$$

Since $D_{ij}(s,t) \geq 0$,

$$\frac{\partial E_{ij}^{(\alpha)}}{\partial \alpha} \leq 0.$$

Thus, increasing the damping parameter cannot increase the exponential attenuation factor.

A second differentiation yields

$$\frac{\partial^2 E_{ij}^{(\alpha)}}{\partial \alpha^2} = D_{ij}^2(s,t) e^{-\alpha D_{ij}(s,t)} \geq 0.$$

The attenuation factor is therefore decreasing and convex as a function of α.

10.2 Sensitivity of the integral component

Let $\mathfrak{Q}_I^{(\alpha)}$ denote the integral component with generalized damping.

Under the established regularity conditions, differentiation under the integral sign gives

$$\frac{\partial \mathfrak{Q}_I^{(\alpha)}}{\partial \alpha} = -\sum_{i,j,k=1}^{n} \int_0^t D_{ij}(s,t) e^{-\alpha D_{ij}(s,t)} \cdot \sin(P_{ijk}(s,t)) H_{ijk}(\mathbf{x}, s) \, \mathrm{d}s.$$

This derivative measures the sensitivity of the integral component to changes in the damping parameter.

Importantly, its sign is not necessarily negative because the oscillatory factor and the trajectory-dependent contribution may change sign.

Therefore, although the exponential attenuation decreases monotonically with α, the complete integral component need not do so.

10.3 Sensitivity bounds

Since

$$D e^{-\alpha D} \leq \frac{1}{e\alpha}$$

for $D \geq 0$ and $\alpha > 0$, we obtain

$$\left| \frac{\partial \mathfrak{Q}_I^{(\alpha)}}{\partial \alpha} \right| \leq \frac{n^3 t M_2 V^2 (1+P)}{e\alpha}.$$

Hence,

$$\left| \frac{\partial \mathfrak{Q}_I^{(\alpha)}}{\partial \alpha} \right| \leq \frac{n^3 T M_2 V^2 (1+P)}{e\alpha}.$$

This provides a quantitative estimate of parameter sensitivity for positive damping parameters.

10.4 Perturbations of the trajectory

Consider two trajectories

$$\mathbf{x}, \mathbf{y} \in \mathcal{X}_T.$$

Define their separation by

$$\varepsilon = \|\mathbf{x} - \mathbf{y}\|_{C^2}.$$

A natural question is whether small changes in the trajectory produce small changes in the functional.

Assume additionally that the coefficient functions and all derivatives required by the functional are locally Lipschitz on a common bounded region containing the images of both trajectories.

Theorem 10.1 (Local Perturbation Stability)**.** *Under the preceding additional assumptions, there exists a finite constant C_T such that*

$$\sup_{t \in [0,T]} |\mathfrak{Q}(\mathbf{x}, t) - \mathfrak{Q}(\mathbf{y}, t)| \leq C_T \|\mathbf{x} - \mathbf{y}\|_{C^2}.$$

Proof. The functional is composed of finitely many products, quotients, compositions, and integrals of trajectory-dependent functions.

On a common bounded region, the relevant coefficient functions and their derivatives are

locally Lipschitz.

Furthermore, all denominators are bounded away from zero by the nonnegative-weight assumptions.

Consequently, the instantaneous factors in each summand satisfy local Lipschitz estimates with respect to the state, velocity, and acceleration.

The accumulated damping and phase terms inherit corresponding estimates because their integration intervals have uniformly bounded length.

The exponential and sine functions are Lipschitz on bounded intervals.

Applying the product and composition estimates to the integral, derivative, and acceleration components yields finite constants C_I, C_D, C_A such that

$$\sup_t |\mathfrak{Q}_I(\mathbf{x}, t) - \mathfrak{Q}_I(\mathbf{y}, t)| \leq C_I \varepsilon,$$

$$\sup_t |\mathfrak{Q}_D(\mathbf{x}, t) - \mathfrak{Q}_D(\mathbf{y}, t)| \leq C_D \varepsilon,$$

and

$$\sup_t |\mathfrak{Q}_A(\mathbf{x}, t) - \mathfrak{Q}_A(\mathbf{y}, t)| \leq C_A \varepsilon.$$

Combining the estimates gives

$$\sup_t |\mathfrak{Q}(\mathbf{x}, t) - \mathfrak{Q}(\mathbf{y}, t)| \leq (C_I + C_D + C_A)\varepsilon.$$

Setting

$$C_T = C_I + C_D + C_A$$

proves the result.

□

10.5 Interpretation of perturbation stability

The preceding estimate establishes a distinction between boundedness and stability.

Boundedness controls the magnitude of the functional along an individual trajectory.

Perturbation stability controls the difference between functional values associated with two nearby trajectories.

This distinction is relevant when studying approximate trajectories, numerical solutions, and parameter-dependent models.

11 Asymptotic Behavior and Limiting Cases

The behavior of Scott's Coherence Functional under limiting regimes provides additional information about its mathematical structure.

We consider several cases in which individual contributions become dominant, negligible, or asymptotically simplified.

11.1 The small-time regime

Consider the limit

$$t \to 0^+.$$

From the integral bound,

$$|\mathfrak{Q}_I(\mathbf{x}, t)| \leq n^3 t M_2 V^2 (1 + P).$$

Therefore,

$$\lim_{t \to 0^+} \mathfrak{Q}_I(\mathbf{x}, t) = 0.$$

Since the remaining two components are continuous in time,

$$\lim_{t\to 0^+} \mathfrak{Q}(\mathbf{x},t) = \mathfrak{Q}_D(\mathbf{x},0) + \mathfrak{Q}_A(\mathbf{x},0).$$

Thus, the integral contribution becomes negligible over sufficiently short observation intervals.

11.2 The strong-damping regime

Suppose the effective damping rate satisfies

$$\frac{\omega_i(\mathbf{x}(\xi))}{1+|\dot{x}_j(\xi)|} \geq \delta$$

for some $\delta > 0$.

Then

$$|E_{ij}(s,t)| \leq e^{-\delta(t-s)}.$$

For an individual summand, suppose

$$|H_{ijk}(\mathbf{x},s)| \leq M_H$$

for a finite constant M_H.

It follows that

$$\begin{aligned}\left|\int_0^t E_{ij}(s,t)\sin\Big(P_{ijk}(s,t)\Big)H_{ijk}(\mathbf{x},s)\,ds\right| &\leq M_H\int_0^t e^{-\delta(t-s)}\,ds \\ &= \frac{M_H}{\delta}\left(1-e^{-\delta t}\right) \\ &\leq \frac{M_H}{\delta}.\end{aligned}$$

Hence, the individual memory contributions remain uniformly controlled under sufficiently strong damping.

11.3 The weak-damping regime

Suppose the damping coefficient is scaled by a parameter α satisfying

$$\alpha \to 0^{+}.$$

Then

$$E_{ij}^{(\alpha)}(s,t) = e^{-\alpha D_{ij}(s,t)} \longrightarrow 1.$$

For a fixed finite observation interval, the convergence is uniform because the accumulated damping quantity is bounded.

Consequently, the integral component approaches the undamped expression

$$\mathfrak{Q}_I^{(0)}(\mathbf{x},t) = \sum_{i,j,k=1}^{n} \int_0^t H_{ijk}(\mathbf{x},s) \cdot \sin(P_{ijk}(s,t))\, \mathrm{d}s.$$

This limiting case isolates the influence of accumulated phase and nonlinear trajectory dependence from exponential attenuation.

11.4 Large-state behavior

Consider the state-dependent factor

$$R(\mathbf{x},s) = \frac{1}{1 + \sum_{m=1}^{n} \chi_m x_m^2(s)}.$$

If at least one positively weighted state coordinate becomes unbounded along a sequence, then

$$R(\mathbf{x},s) \to 0.$$

For bounded velocities and bounded remaining integrand factors, the corresponding velocity contribution therefore approaches zero.

This identifies a conditional attenuation mechanism associated with large weighted state magnitudes.

11.5 Limiting behavior of the acceleration term

Consider the scalar expression

$$F(u, v) = \frac{uv}{1 + |u| + |v|}.$$

If u is fixed and $v \to 0$, then

$$\lim_{v \to 0} F(u, v) = 0.$$

Likewise, if v is fixed and $u \to 0$,

$$\lim_{u \to 0} F(u, v) = 0.$$

If both quantities become large with comparable magnitudes, the expression can also become large.

Thus, the nonlinear denominator moderates the growth of the acceleration contribution but does not establish universal boundedness.

12 Special Cases and Simplifications

The general expression contains several mechanisms that can be examined independently through particular parameter choices.

12.1 Vanishing acceleration contribution

If

$$\Lambda_{ijk} = 0$$

for every triple of indices, then

$$\mathfrak{Q}_A = 0.$$

The functional reduces to

$$\mathfrak{Q} = \mathfrak{Q}_I + \mathfrak{Q}_D.$$

12.2 Vanishing state weights

If

$$\chi_m = 0$$

for every m, then

$$1 + \sum_{m=1}^{n} \chi_m x_m^2 = 1.$$

The velocity contribution simplifies to

$$\dot{x}_j(s)\dot{x}_k(s).$$

12.3 Vanishing oscillatory phase

If

$$\Omega_{ijk} = 0$$

and

$$\Theta_{ijk} = 0,$$

then

$$\sin(0) = 0.$$

Consequently,

$$\mathfrak{Q}_I = 0.$$

The functional then consists of its derivative and acceleration contributions.

12.4 Constant trajectories

Suppose

$$\mathbf{x}(t) = \mathbf{x}_0$$

for every t.

Then

$$\dot{x}_i(t) = 0$$

and

$$\ddot{x}_i(t) = 0.$$

The integral contribution vanishes because its numerator contains a product of velocities.

The acceleration contribution also vanishes.

Therefore,

$$\mathfrak{Q}(\mathbf{x}_0, t) = \sum_{i=1}^{n} \frac{\partial^3 \Phi_i}{\partial x_i^2 \partial t}(\mathbf{x}_0, t) \cdot \prod_{j=1}^{n} (1 + \kappa_{ij} x_{0,j}^2).$$

This special case isolates the derivative contribution of the functional.

12.5 One-dimensional reduction

For $n = 1$, all triple summations collapse into a single term.

Writing $x = x_1$, the functional becomes

$$\begin{aligned}\mathfrak{Q}(x,t) = \int_0^t & \Phi_{1,xx}(x(s),s)E(s,t) \\ & \cdot \sin(P(s,t))\frac{\dot{x}^2(s)}{1+\chi_1 x^2(s)} \\ & \cdot \left(1 + \frac{\Psi_{1,x}(x(s),s)}{1+|\Psi_{1,x}(x(s),s)|}\right) \mathrm{d}s \\ & + \Phi_{1,xxt}(x(t),t)(1+\kappa_{11}x^2(t)) \\ & + \Lambda_{111}\frac{\dot{x}^2(t)\ddot{x}(t)}{1+\dot{x}^2(t)+|\ddot{x}(t)|}.\end{aligned}$$

This reduction provides a simpler setting for numerical experimentation and analytical examination.

13 Possible Generalizations

The functional can be generalized in several directions without changing its fundamental three-component structure.

13.1 Generalized damping

Introduce a positive parameter α:

$$E_{ij}^{(\alpha)}(s,t) = \exp\left(-\alpha \int_s^t \frac{\omega_i(\mathbf{x}(\xi))}{1+|\dot{x}_j(\xi)|}\,\mathrm{d}\xi\right).$$

The parameter controls the strength of the damping contribution.

For $\alpha = 1$, the original expression is recovered.

13.2 Generalized oscillatory functions

The sine function may be replaced by a more general bounded continuous function $g : \mathbb{R} \to \mathbb{R}$.

The corresponding factor becomes

$$g(P_{ijk}(s,t)).$$

If

$$\sup_{\theta \in \mathbb{R}} |g(\theta)| < \infty,$$

then the principal existence and boundedness arguments remain applicable.

13.3 Higher-order trajectory dependence

The acceleration term may be extended to include third-order derivatives.

For example,

$$\mathfrak{Q}_J = \sum_{i,j,k=1}^{n} \Gamma_{ijk} \frac{\dot{x}_i \dot{x}_j \dddot{x}_k}{1 + |\dot{x}_i \dot{x}_j| + |\dddot{x}_k|}.$$

Such an extension requires trajectories with sufficient additional regularity, for example

$$\mathbf{x} \in C^3([0,T], \mathbb{R}^n).$$

13.4 Alternative nonlinear attenuation

The state-dependent denominator may be generalized to

$$1 + \sum_{m=1}^{n} \chi_m |x_m|^p, \qquad p > 0.$$

For nonnegative coefficients, this expression remains positive.

The parameter p controls the rate at which the attenuation changes with state magnitude.

13.5 Generalized product weighting

The derivative contribution may also be extended through a positive weighting function w_{ij}:

$$\mathfrak{Q}_D^{(w)} = \sum_{i=1}^{n} \frac{\partial^3 \Phi_i}{\partial x_i^2 \partial t} \prod_{j=1}^{n} w_{ij}(x_j).$$

Suitable assumptions on w_{ij} would preserve the existence of this component.

These generalizations suggest that the original construction can be regarded as one member of a broader family of trajectory-dependent nonlinear functionals.

14 Limitations and Open Mathematical Problems

Although the preceding analysis establishes several fundamental properties of Scott's Coherence Functional, a number of mathematical questions remain unresolved.

The distinction between established results and potential extensions is important for understanding the actual scope of the construction.

14.1 Trajectory-independent bounds

The global estimate established in this paper depends on

$$X, \qquad V, \qquad A.$$

These quantities represent bounds on the trajectory, velocity, and acceleration.

Consequently, the existing estimate does not establish a universal upper bound over the entire space $\mathcal{X}_T$.

A natural extension would investigate whether additional restrictions on admissible trajectories could produce estimates independent of individual trajectory magnitudes.

14.2 Differentiability with respect to trajectories

Continuity with respect to trajectories does not automatically imply differentiability.

A further investigation would examine whether the mapping

$$\mathbf{x} \mapsto \mathfrak{Q}(\mathbf{x}, t)$$

possesses a Fréchet derivative on appropriate subsets of the trajectory space.

If such a derivative exists, it would provide a linear approximation of the functional under small perturbations.

Specifically, one would seek a bounded linear operator $D\mathfrak{Q}(\mathbf{x})$ satisfying

$$\begin{aligned}\mathfrak{Q}(\mathbf{x}+\mathbf{h}, t) = \mathfrak{Q}(\mathbf{x}, t) \\ + D\mathfrak{Q}(\mathbf{x})[\mathbf{h}] + o(\|\mathbf{h}\|_{C^2}),\end{aligned}$$

as

$$\|\mathbf{h}\|_{C^2} \to 0.$$

Establishing this property would provide a foundation for a more complete variational analysis.

14.3 Existence of conservation laws

The current formulation does not establish that the functional is conserved along solutions of a particular differential equation.

A conservation property would require additional structural assumptions.

One possible objective would be to identify conditions under which

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathfrak{Q}(\mathbf{x}, t) = 0$$

along a specified class of trajectories.

Such a result would require an appropriate evolution equation and a detailed analysis of the time derivative of each component.

14.4 Monotonicity and sign structure

The oscillatory integral component is not necessarily nonnegative.

Indeed, the sine factor can change sign, while the derivative and acceleration contributions may also be positive or negative.

Therefore, the functional cannot generally be interpreted as a positive-definite quantity.

An interesting open problem is to identify additional restrictions on the coefficient functions and trajectories that guarantee monotonicity or sign definiteness.

14.5 Numerical approximation

Another direction concerns the numerical evaluation of the functional.

The nested integrals and nonlinear derivative terms introduce several potential sources of numerical error.

A numerical investigation could compare quadrature methods and trajectory approximation schemes.

One possible objective would be to establish an estimate of the form

$$|\mathfrak{Q}(\mathbf{x}, t) - \mathfrak{Q}_h(\mathbf{x}, t)| \leq Ch^p,$$

where h denotes a discretization parameter and p represents the convergence order.

The derivation of such an estimate would require assumptions on the smoothness of the integrand and the selected numerical approximation method.

14.6 Existence of stationary trajectories

Another possible direction is the identification of trajectories for which the functional becomes stationary.

A corresponding investigation could seek trajectories satisfying

$$D\mathfrak{Q}(\mathbf{x})[\mathbf{h}] = 0$$

for every admissible perturbation $\mathbf{h}$.

This would require an appropriate differentiability theory and additional conditions on the trajectory space.

These unresolved questions provide several directions for subsequent mathematical investigation.

15 Conclusion

This paper introduced Scott's Coherence Functional, a trajectory-dependent mathematical construction combining integral, derivative, oscillatory, damping, and acceleration contributions.

The functional contains three principal components.

The first combines mixed second-order spatial derivatives with accumulated exponential damping, oscillatory phase, and nonlinear velocity dependence.

The second introduces third-order derivatives weighted by products of state-dependent factors.

The third introduces a nonlinear correction involving velocity and acceleration.

Under appropriate regularity, boundedness, and sign assumptions, the functional was shown to be well-defined on finite time intervals.

An explicit global bound was established in terms of the trajectory, velocity, acceleration, and relevant coefficient functions.

Continuity in time was also demonstrated, allowing the functional to attain maximum and minimum values on compact intervals.

The analysis further identified a nonlocal memory structure within the integral component.

Under a uniformly positive effective damping rate, the total absolute memory weight admits a uniform bound independent of the observation time.

Parameter sensitivity was examined through a generalized damping parameter, and additional assumptions were used to establish local perturbation stability with respect to trajectory changes.

Several limiting cases were also investigated, including short-time, strong-damping, weak-damping, and large-state regimes.

The construction admits potential generalizations involving alternative oscillatory functions, higher-order trajectory derivatives, and nonlinear attenuation mechanisms.

Nevertheless, several mathematical questions remain open.

These include Fréchet differentiability, trajectory-independent bounds, conservation properties, sign definiteness, and numerical convergence.

A particularly interesting direction is the development of a complete variational theory for $\mathfrak{Q}$, examining how changes in the trajectory influence the resulting functional.

The present work provides an initial analytical foundation for these investigations and establishes a framework within which the structural properties of Scott's Coherence Functional can be studied more systematically.

16 Acknowledgments

The author thanks the teachers and mentors who encouraged independent mathematical exploration and the development of original mathematical ideas.

The author also acknowledges the foundational contributions of researchers in functional analysis, differential equations, and mathematical analysis, whose established results provide important tools for the present work.

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